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Recovery-Aware Physical Event Attribution From Sparse PMUs

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ABSTRACT An electrical network can continue responding after an intervention is restored, while a new event begins before that response decays. With sparse PMUs, this creates ambiguity between a new source and the recovery of an earlier one. We formulate attribution through competing physical histories. The inference engine uses switched state-space predictors derived from the network differential-algebraic equations, together with voltage, terminal current, frequency and ROCOF from eight PMUs. It screens unknown onsets over a ten-second retrospective window, integrates load severity, and preserves separate active-event and originating-cause masses after restoration. A finite bank represents constant network-parameter uncertainty. In completed development streams, the bank obtains 53/54 timely family/source decisions with 4 unmatched alarms. Nominal gains one timely decision but produces more extras. The independent, prespecified evaluation remains in progress. A separate known-onset load experiment illustrates why visible weak interventions can remain difficult to localize. The contribution is a specified recovery-aware attribution mechanism and an evaluation that separates detection, timely source decisions and restoration. The validated scope is a known load/line catalog; calibrated event probabilities, all-family diagnosis and real-time execution are not established.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AN unmeasured load changes. Several PMUs respond, a detector raises an alarm, and the network keeps oscillating after the load is restored. A second intervention then begins during that recovery. The operational question is no longer whether one signal exceeds a threshold: which physical intervention explains the reports observed so far, where did it originate, and when is that explanation distinguishable from the alternatives?

Sparse PMUs make the question difficult because a source need not have a sensor at its own bus. Its voltage, current and frequency effects propagate through the network [1], [2]. Uncertain operating context and network parameters can also change those effects. Moreover, physical OFF is not electrical recovery. A method that resets at restoration can treat the remaining response as another event or lose a real event that begins before the oscillation decays.

Strong existing methods solve parts of this problem. Dynamic state estimators combine power-system equations and PMU histories [3]; supervised classifiers learn sparse-PMU event/source labels [4], [5]; physical outage detectors compare network hypotheses sequentially [6], [7]; and high-fidelity fault trajectories already support sparse-PMU diagnosis [8]. The question here is how to retain an explanation through restoration and compare its recovery with a later intervention. A low state error does not prove source identity, and a score on represented assets does not prove source transfer.

Our approach uses the physical intervention as the object of inference. DAE derivatives construct switched predictors for load and line candidates. Starting from an estimated past state, those predictors describe what the eight PMUs should report. Conditional likelihoods compare the predictions with observations while accounting for severity and instrumental memory. A finite network-model bank represents parameter uncertainty. Active and originating-cause masses distinguish an ongoing intervention from the response that remains after restoration.

The paper makes three scoped contributions:

1) **Recovery-aware physical attribution:** an executable

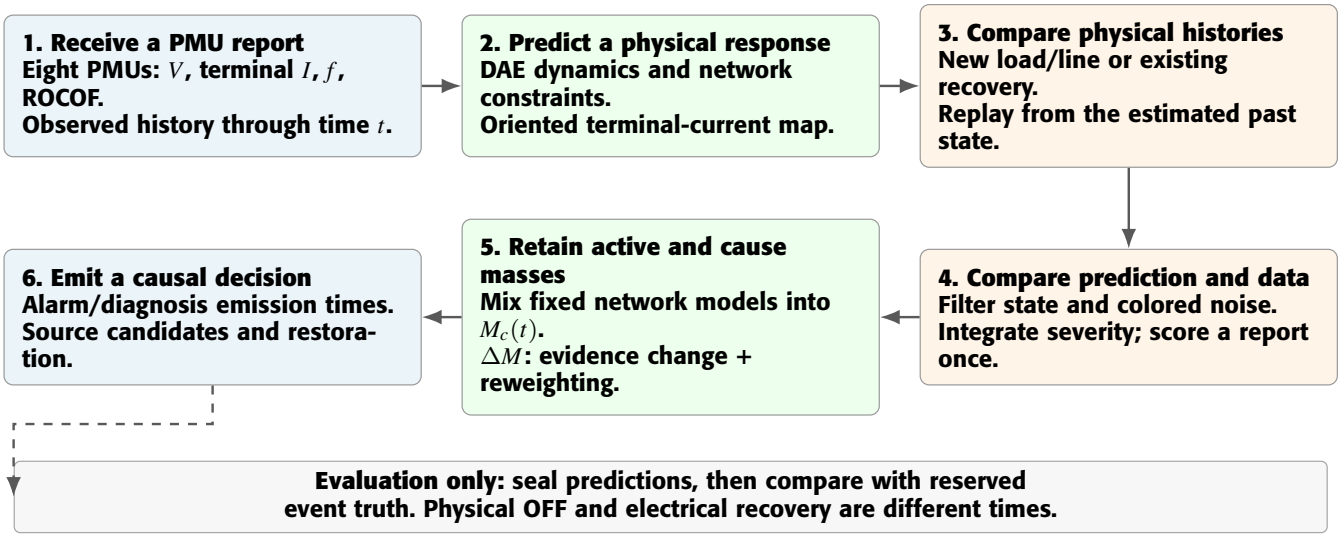


FIGURE 1. **The complete scientific argument. Physics maps an intervention at an unmeasured source into predicted PMU trajectories; causal evidence chooses among those explanations. Event truth enters only evaluation. The figure describes the implemented restricted load/line component, not an already validated all-family joint estimator.**

causal load/line history bank starts from estimated past states, retains the originating cause after physical OFF, and compares recovery with new-intervention histories under fixed network hypotheses. We specify its evidence and publication rules.

2) **A qualified source-resolution analysis:** a separate archived load-support study exposes ambiguity and the different information used by joint and conditional source decisions. A prior-consistency audit distinguishes implementation effects from a claimed physical limit; this study is not counted as causal accuracy.
3) **A reproducible operational test:** the nominal/mixture comparison and a failed admission ablation expose the trade-off between timely source attribution and extra alarms. Separate state, batch and stream contracts retain their own denominators. R11 and nominal remain frozen co-references for confirmation.

The central question is whether this recovery-aware construction preserves timely source decisions without excessive extra alarms. The experiments separate development from independent evaluation and keep detection, eventual attribution and attribution before restoration as different outcomes. Physical templates, onset scanning and Bayesian updates have close precedents; the claimed contribution is their specified operational construction for the declared load/line problem and its measured limits.

II. CLOSEST METHODS AND THE REMAINING QUESTION

Dynamic state estimation supplies the state and uncertainty machinery used to predict a PMU history. Unknown-input and DAE estimators address incomplete models and exogenous inputs [3], [9], [10]. Their task is not automatically an event-family/source decision. A recovered bus-voltage field can remain compatible with competing physical interventions.

Model-based source inference is the closest methodological precedent. Phasor angles support outage hypotheses [11]; uncertain pre-event conditions can be marginalized with outage identity [12]; and sequential likelihoods detect line changes under stochastic operation [6], [7]. Anguluri *et al.* jointly recover sparse forcing locations, inputs and an unknown initial state in linear systems using group LASSO [13]. That is a direct precedent for physical source recovery, although forced inputs and linear dynamics differ from the switched load/line episode task here. It is not an executed C01 comparator. Transient dynamics, uncertain initial states and unknown onset are therefore established ingredients, not contributions of this paper.

Warner *et al.* provide a particularly close precedent [8]. Their simulated voltage/current fault trajectories are compared with a steady-state reference over a moving onset window; Algorithm 1 stops and returns candidate lines. Load changes are an adverse-event test rather than a load-source output. Their Sections II and IV target short-circuit identification before clearing, whereas our restricted task follows load changes and line openings through restoration and later interventions. Sparse sensing, a physical trajectory library and retrospective onset scanning are shared ideas. We have not run their procedure on the present data and make no numerical superiority claim over it.

Sequential diagnosis also has a stronger statistical literature than a rolling-window label suggests. The Adaptive Matrix CuSum addresses misidentification caused by pre-change observations and establishes delay/error results under its specified observation laws [14]. R11's pruned conditional histories, colored measurement state and plug-in startup estimates do not inherit those guarantees. A ten-second window or the array M is not a replacement for that theory.

Learned PMU diagnosis provides a different comparator. Sparse classifiers and knowledge-driven representations identify represented events or assets [4], [5], [15]. The historical SGSMA baseline supplies two frozen ExtraTrees adaptations here. We distinguish interpolation at represented sources, transfer of that frozen adapter, and true held-source generalization. The present causal catalog is known; its results do not establish the last of those claims.

Observability and diagnosability explain why neither a physical nor learned method is guaranteed to localize every visible event. Functional observability concerns the requested state function [16], [17]; descriptor-model diagnosability and nuisance-profiled tests concern competing explanations [18], [19]. Placement-dependent source ambiguity is also established in PMU fault/event work [20]–[22]. Our state/source distinction applies these ideas; it is not their first discovery.

Multiple-model filtering already manages conditional state/mode hypotheses [23]; R11 is a pruned history bank, not a new IMM derivation. Stochastic swing/kernel models estimate latent states in Ensaf and Barati [24], and physics-derived prior covariances support sparse-state forecasting [25]. Joint DAE-aware generator/network parameter inference is studied separately [26]. R11 uses a finite bank of constant network hypotheses rather than inferring the continuous parameter posterior. A Gaussian process or neural architecture would require separate validation.

The remaining question is operational and falsifiable: can a state-conditioned bank distinguish a new intervention from a preceding intervention's recovery while retaining timely family/source decisions under fixed network uncertainty? Our contribution is the specified recovery-aware construction and its bounded evidence, not a claim to have invented model-based diagnosis. We use "physics-informed" because physical equations define candidate trajectories and likelihood inputs. Learned discrepancy and the unrestricted joint state/event/integrity estimator remain separate proposals.

III. THE DECISION REQUIRED FROM EIGHT PMUS

At every report, the task is to decide whether the recent trajectory contains a physical intervention and, if so, identify its family, source and onset. Detection is the existence decision; classification chooses the physical operator; localization chooses a bus or branch. Those outputs need not become reliable at the same time. We keep the estimated physical onset $\widehat{\tau}$ separate from the time at which an alarm or diagnosis is emitted.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine, archived as R11, uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents

TABLE 1. Closest inference contracts and the difference tested here. This is a methodological comparison; only Chen and the frozen ExtraTrees adapters are executed comparators in C01. Absence from a paper's evaluated task is not a claim of algorithmic impossibility.

Route	Physical/statistical mechanism	Published target	Difference or role here
Chen / Rovatsos	Increment statistics, network models and sequential tests	Line outages; Rovatsos includes transient phases	Chen is executed on persistent-line first stops
Warner high-fidelity	Simulated V/I paths, moving onset window, distance evidence	Short-circuit line candidates	Nearest trajectory precedent; no common-data run here
Adaptive Matrix CuSum	Pairwise sequential diagnosis and error/delay analysis	One unknown change among distributional alternatives	No corresponding optimality guarantee is claimed here
DAE / stochastic-state estimation	Incomplete DAE observers or physical uncertainty models	States, unknown inputs or network parameters	Components/adjacent tasks, not episode attribution scores
Knowledge-driven learned diagnosis	Spatiotemporal representation with topology/type constraints	Multi-family disturbance location	Distinct from the physical likelihood; ExtraTrees is our executed learned reference
Physical history bank here	Estimated-past-state paths, active/cause masses, fixed R/X mixture	Successive known-catalog load/line interventions	Test whether recovery-aware attribution retains timely sources without excess alarms

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Observation derivatives provide the corresponding reduced output. The implementation discretizes these operators for the admitted switch modes. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{7}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{8}$$

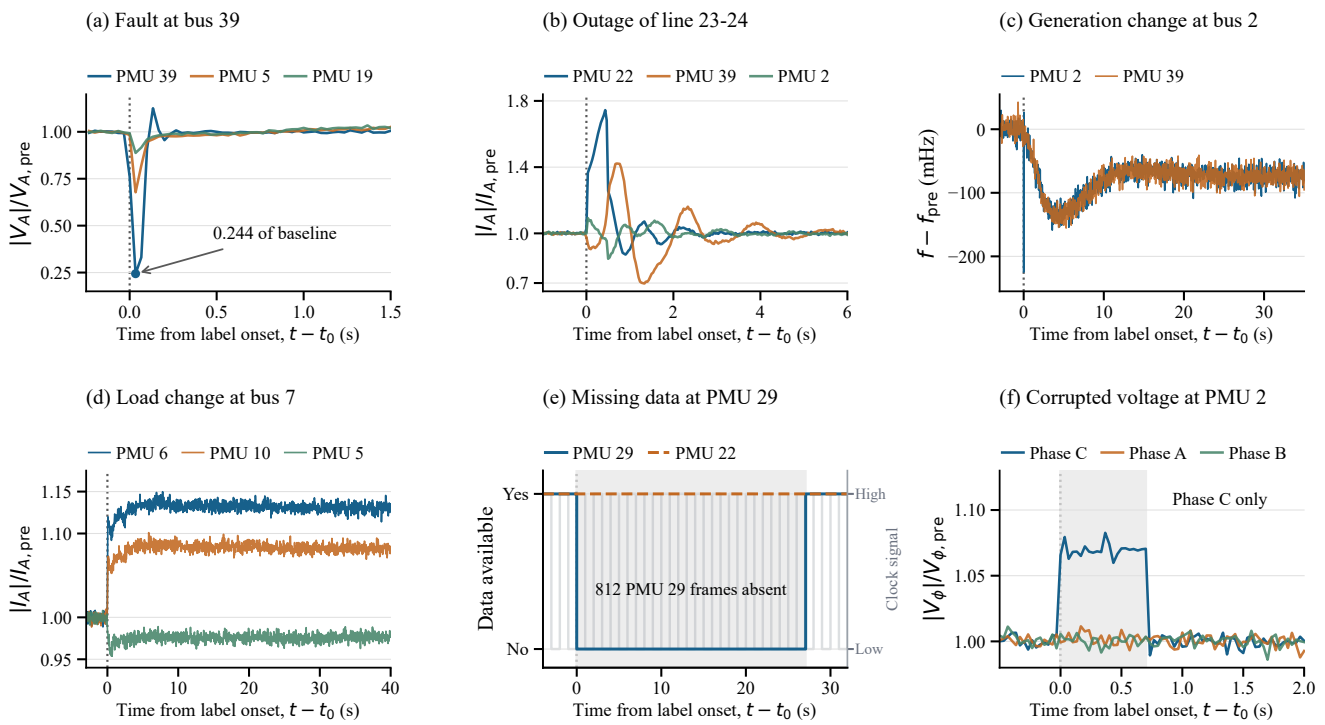


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

At time t_k, the proposal screen considers onsets in $[t_k - 10\,\mathrm{s}, t_k]$. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{9}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{10}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = {} & \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ & + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{11}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{12}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

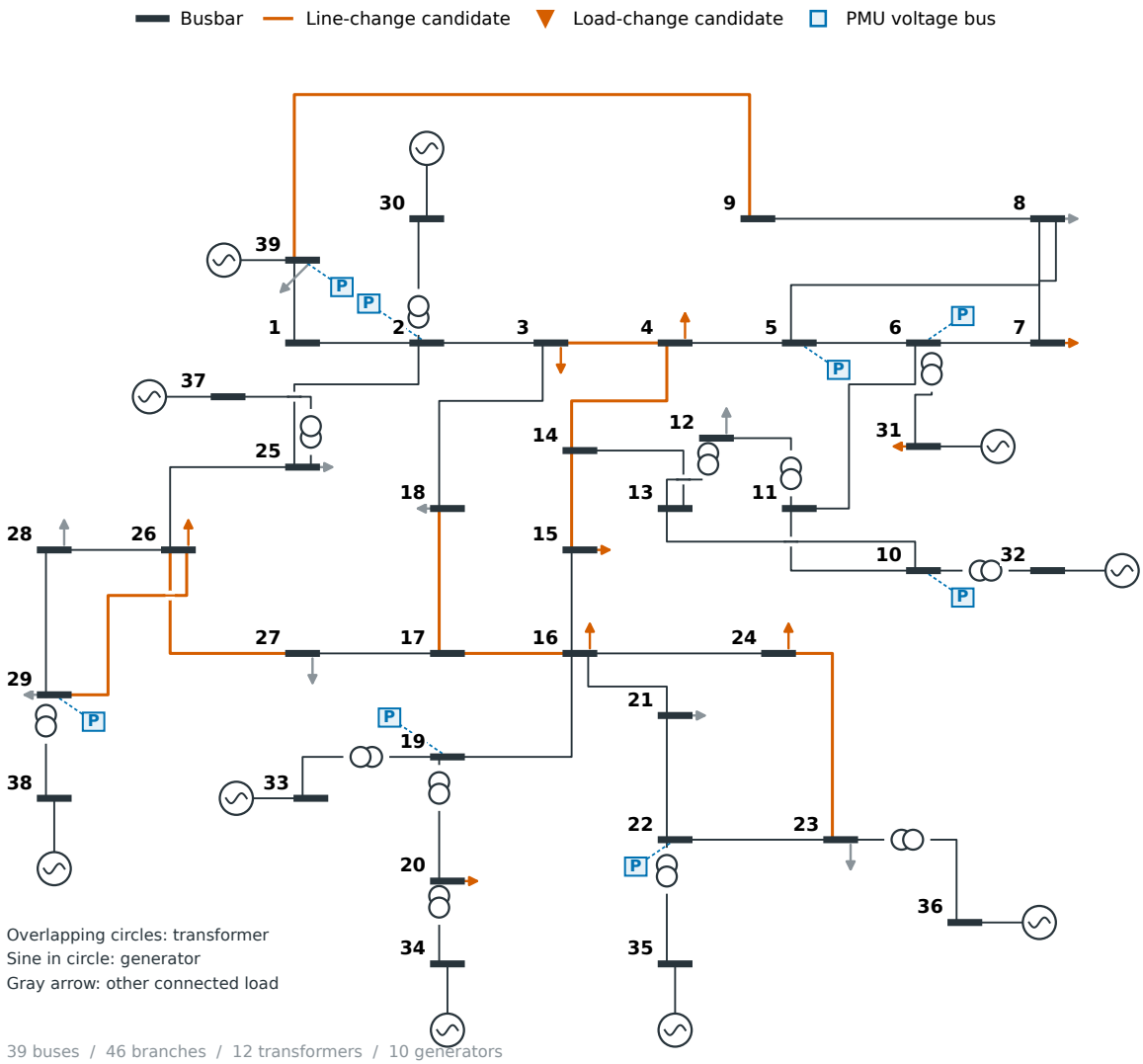


FIGURE 3. **Electrical single-line diagram of the frozen IEEE-39 model: 39 busbars, 46 branches, 12 transformers (overlapping circles), and 10 generators (sine symbols). Arrows denote connected loads; orange arrows and lines mark the nine load-change buses and nine candidate line interventions. Blue P badges identify the eight voltage-measurement buses; current channels retain the separate oriented terminal assignments described in the measurement contract. Unconnected wire crossings have gaps. The drawing is schematic, not a geographic or electrical-distance map.**

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{13}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{14}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus.

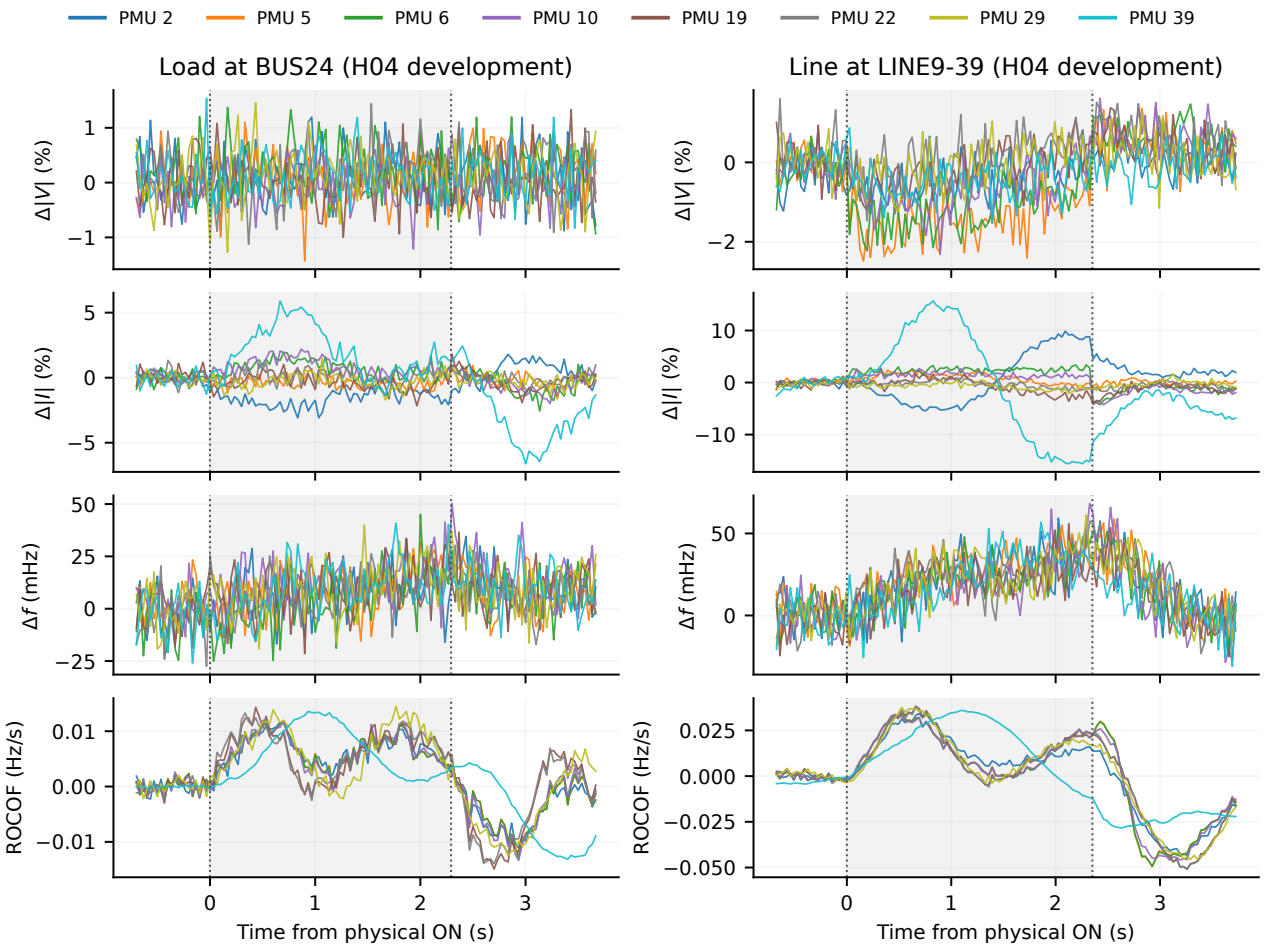


FIGURE 4. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \quad (15)$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture tests the outer network bank. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report.

3) Screen permitted family/source/onset proposals in the preceding ten seconds. A proposed history starts from its retained past state; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score the report using its predictive density, before correcting the state with that report. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The simulator is the IEEE-39 Julia/PowerDynamics research candidate F01. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on H04 or C01. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared F01 adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign, C01, therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent evaluation is in progress. The following section reports completed development results only.

TABLE 3. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Auxiliary load-resolution study	137 supports; known onset and context	Cardinality/support accuracy by severity	Different batch estimator; separate from causal accuracy

VI. CONTINUOUS-STREAM RESULTS

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 4). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. Representing model uncertainty trades an additional late diagnosis for fewer extra alarms here.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 6 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (15) separates that revision from within-model evidence. This is what the array adds to the explanation. The figure does not establish that a tensor representation alone improves accuracy.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative experiment leaves R11 and nominal frozen for C01.

TABLE 4. What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

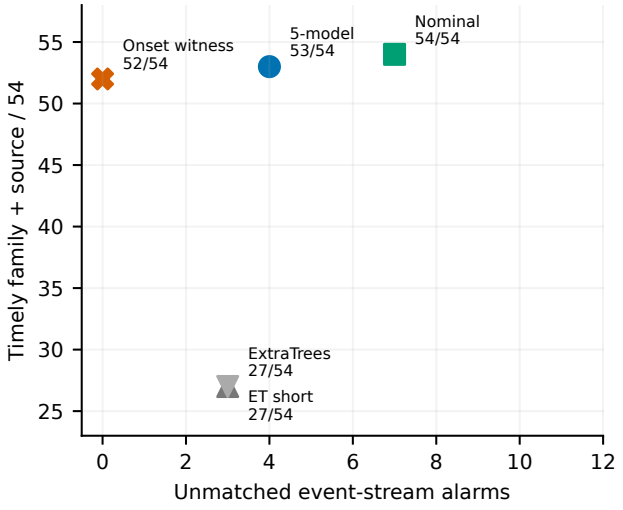


FIGURE 5. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

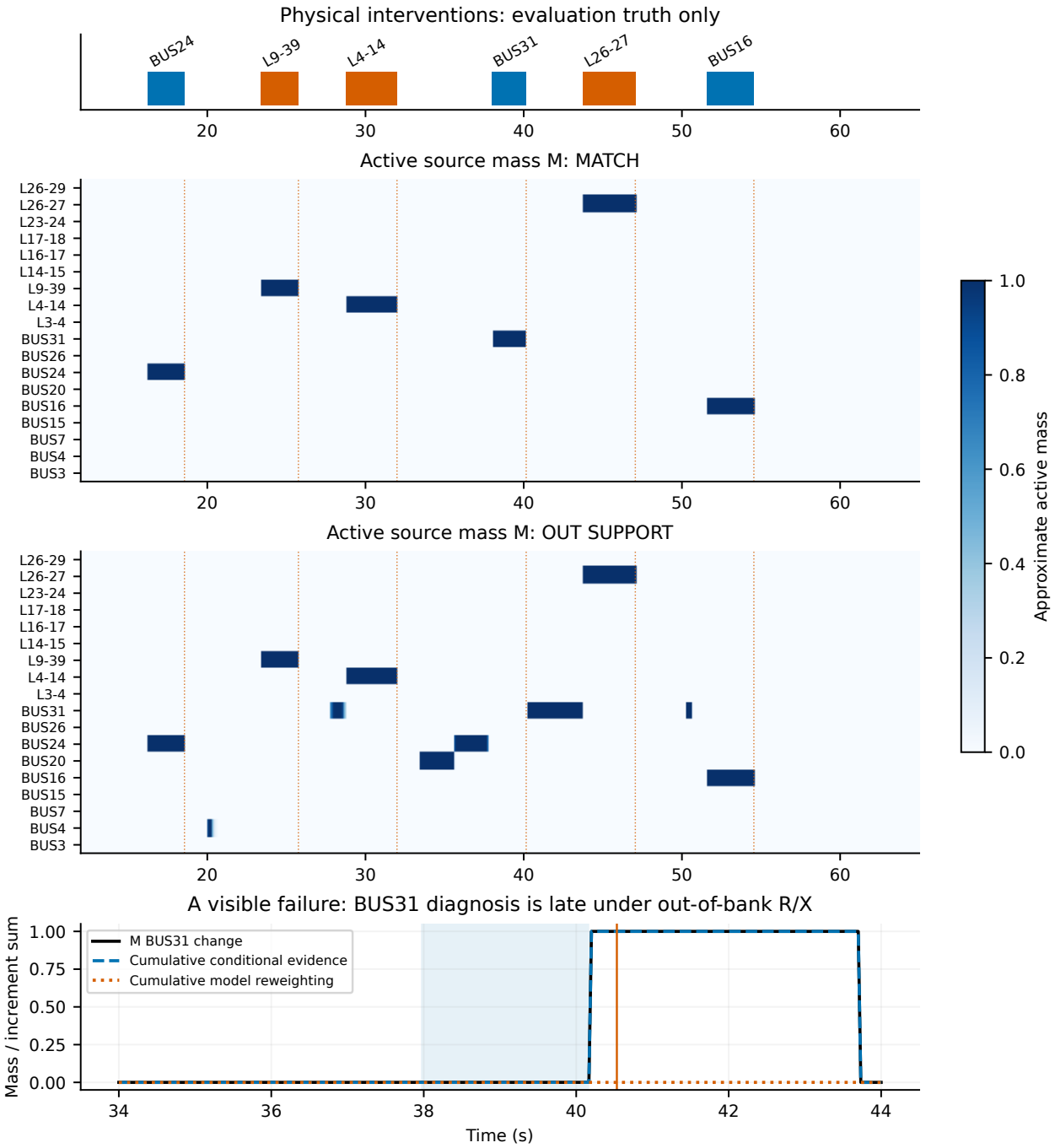


FIGURE 6. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

VII. WHY VISIBLE EVENTS CAN HAVE AMBIGUOUS SOURCES

The continuous results raise a separate physical question: can these measurements distinguish the candidate sources at all? A good state estimate does not answer it. Different interventions can produce similar histories after fitting their severity and initial state [18], [19].

For a response s_h at a specified severity and context, its distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\eta_j \in \mathcal{A}_j} \|s_h - s_j(\eta_j)\|^2_{\Sigma^{-1}}. \tag{16}$$

The set $\mathcal{A}_j$ specifies allowed severity, sign and nuisance coordinates. If zero severity is admitted, different support families can meet at the no-event boundary. Unrestricted amplitude minimization would then give a misleading source-separation interpretation. Under a common Gaussian law, coincident means cannot be distinguished at that intersection. Small distances indicate a noise-sensitive problem, not a universal error formula or an online uncertainty certificate.

A separate archived load-only experiment illustrates observed ambiguity. It uses 32 rectangular voltage/current coordinates, known onset and known operating context. Its original amplitude-integrated batch score compares 137 zero-, one- and two-source supports on 3488 fresh nonlinear trajectories. It is a different estimator and evidence contract from the causal bank. Its scores require the normalization qualification below before being interpreted under the originally declared priors.

Exact-support Top-1 increases from 24.4% for weak pairs to 100.0% for finite pairs (Fig. 7). A visible intervention therefore need not yield a reliable location in this implementation. Top-3 in the archive ranks only pair supports, whereas Top-1 also requires the full bank to select two sources. They must not be compared as though only the number of returned labels changed. These observed errors do not separate insufficient information from model, prior or numerical error; that separation requires an exact-response control.

The archived implementation also requires a prior qualification. Its single-source integral carries an extra factor $1/(2\pi)$ relative to the null and pair integrals, giving an effective cardinality prior proportional to $(0.20, 0.50/(2\pi), 0.30)$ instead of the originally declared $(0.20, 0.50, 0.30)$. A zero-response control on the historical code exposed this difference. An isolated successor restores the declared prior and passes independent linear-Gaussian checks for all 136 event supports; its physical performance has not been recomputed. Figure 7 retains the archived implementation's outcomes. This issue does not alter the R11 integral in (11), checked separately by numerical quadrature.

Longer history and different PMUs may help discrimination, but these curves do not certify an online uncertainty or abstention rule. Exact replay of the auxiliary study requires restoring source arrays absent from this artifact copy; no newly fitted dictionary replaces them. Detailed state-reconstruction studies remain supporting evidence rather than extra event-diagnosis scores.

VIII. INTERPRETATION AND LIMITATIONS

The physical bank explains a PMU history through interventions, states and network hypotheses. Its distinguishing operational choice is to retain a restored intervention as a possible cause of the continuing response. This makes recovery compete with a later event instead of treating every departure from a flat baseline as new. The nominal/mixture comparison shows a trade-off in development: fewer extras can accompany a late source decision. Stricter admission eliminates extras but can also reject a real load. These observations support reporting timely attribution and alarm burden together.

This construction builds on established trajectory-based diagnosis, multiple-model filtering and sequential inference. Warner's physical trajectory scan already handles unknown onset and sparse remote PMUs; Adaptive Matrix CuSum provides error/delay theory for its statistical setting. We do not inherit those guarantees or claim numerical superiority over methods not executed on common data. The present admission ablation also does not isolate the causal benefit of every recovery-state operation.

The executed scope is a known catalog of nine loads and nine lines, with an intervention-free startup. Missing PMUs, unknown event families, source-excluded transfer and field operation are not tested. F01's measurement map includes the provisional BUS5 current terminal and excludes BUS39 current angle. Agreement with this research model does not establish the original producer's CT placement or exact replication of RAW0001. Local predictors, pruning, shared covariance approximations and plug-in startup estimates limit the interpretation of the masses as probabilities.

The ExtraTrees comparison concerns two specific fixed causal adapters. Their load-transfer failure is not evidence that supervised learning is incapable of recognizing loads, and its score cannot be substituted for a historical multi-family benchmark. Chen's F01 adapter has DC/white-injection assumptions that do not capture all instrumental effects; its failures must be reported with that mismatch.

Physics-derived stochastic priors offer a coherent direction for future prediction uncertainty [24], [25], while joint DAE parameter estimation addresses another task [26]. Such extensions would require separate validation. They are not components silently added to the method evaluated here. The current scientific claim rests on the specified recovery-aware mechanism and the evidence produced by its fixed experimental contract.

A specific successor question is whether context should change the event signature itself. For pre-event context η, a mixed sensitivity $E_i = \partial D_i/\partial \eta$ changes the predicted effect of source i, rather than merely shifting the nominal trajectory. Under a local Gaussian context approximation, its predictive covariance would be

$$V_h(a) = \Sigma_W + J_{\eta,h}(a) P_{\eta,\tau} J_{\eta,h}(a)^\top. \tag{17}$$

The determinant of $V_h(a)$ must remain in the evidence. Testing this extension requires matched recentering-only and

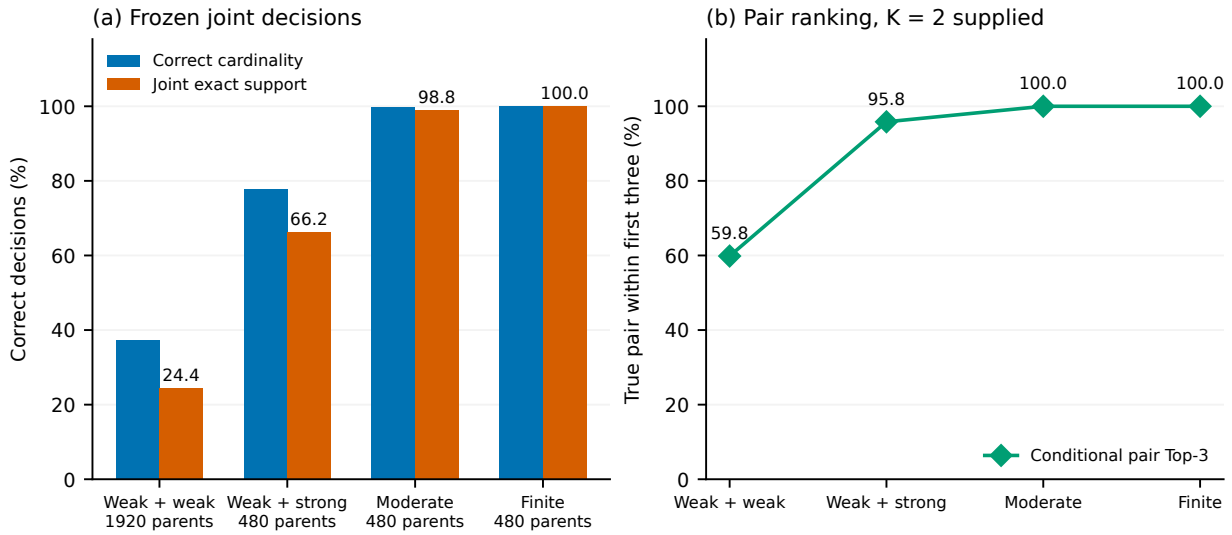


FIGURE 7. Archived load-support outcomes, not causal R11 accuracy. Left: cardinality and joint exact-source decisions of the frozen 137-support implementation. Right: Top-3 rank restricted to the 120 pairs, conditional on knowing that two sources occurred; it is a different decision contract. Labels give independent physical trajectories, each replayed with three noise seeds. The evidence-normalization audit below qualifies the cardinality prior; these curves are not corrected-prior results or proof of an information-theoretic limit.

mixed-sensitivity controls, calibrated pre-event uncertainty and new physical trajectories. The present R11 results do not validate that joint multiple-source algorithm.

A decisive test would hold observations, priors and horizon fixed when comparing nominal, recentered and context-conditioned signatures. Varying horizon separately would identify gains from additional data. Nonlinear-response controls would distinguish surrogate error from confusion among physical alternatives. These successor experiments must measure source decisions and uncertainty calibration; algebraic equivalence or lower prediction error alone cannot establish either benefit.

IX. CONCLUSION

Restoring an operator does not erase the dynamic state it created. We used that physical distinction to construct a causal attribution bank for sparse PMUs: load/line histories start from estimated past states, retain their originating cause during recovery, and compete under fixed network hypotheses. M summarizes active-source mass; ΔM separates within-model evolution from network reweighting.

Completed development streams give 53/54 timely family/source decisions with 4 extras for the mixture. Nominal gains one timely decision at a higher alarm burden. Independent confirmation remains in progress. The auxiliary archive motivates a complementary question: how much source ambiguity comes from the observations, approximation and evidence implementation? Its normalization audit prevents treating an archived score as a certified physical limit. The continuous results support a restricted study of successive load/line attribution, rather than an all-family or universally optimal estimator.

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REFERENCES

[1] A. G. Phadke and J. S. Thorp, *Synchronized Phasor Measurements and Their Applications*. New York, NY, USA: Springer, 2008.

[2] J. Zhao, J. Qi, Z. Huang, A. P. S. Meliopoulos, A. Gómez-Expósito, M. Netto *et al.*, "Power system dynamic state estimation: Motivations, definitions, methodologies and future work," *IEEE Transactions on Power Systems*, vol. 34, no. 4, pp. 3188–3198, 2019.

[3] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[4] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.

[5] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.

[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023, posted June 16, 2023; cited as a preprint.

[9] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[10] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[11] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.
[12] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.
[13] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.
[14] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[15] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[16] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.
[17] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.
[18] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
[19] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
[20] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
[21] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
[22] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
[23] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.
[24] M. Ensaf and M. Barati, "Harnessing kernel regression for stochastic state estimation in solar-integrated power grids," arXiv preprint arXiv:2309.10200, 2023, version 1. [Online]. Available: https://arxiv.org/abs/2309.10200
[25] A. M. Tartakovsky, T. Ma, D. A. Barajas-Solano, and R. Tipireddy, "Physics-informed gaussian process regression for states estimation and forecasting in power grids," *International Journal of Forecasting*, vol. 39, no. 2, pp. 967–980, 2023.
[26] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," arXiv preprint arXiv:2604.15686, 2026, preprint. [Online]. Available: https://arxiv.org/abs/2604.15686